

An Acoustic and perceptual study of Japanese voiced geminates*

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1. Introduction

It is well-known that keeping voicing in obstruents is articulatorily challenging. With obstruent closure, the intraoral air pressure goes up very quickly, and as a consequence a sufficient transglottal air pressure drop to keep voicing becomes harder to obtain. This difficulty becomes more problematic in geminate obstruents as they have long closure (see Hayes and Steriade in press; Jaeger 1978; Kingston 2003; Kirchner 1998; Ohala 1983). Reflecting this articulatory difficulty of maintaining voicing in long obstruent closure, Japanese did not have a voicing contrast in geminates i.e., no voiced geminates were allowed. In addition to distributional evidence, various strategies such as coda nasalization in mimetics gemination shown in (1) (Kuroda 1965; Ito and Mester 1999) or blockage of syncope in Sino-Japanese (Ito and Mester 1996; Kurisu 2001) shown in (2) conspire to avoid having voiced geminates in the non-loanword phonology of Japanese:

- | | | | | | |
|-----|----------------|---|------------|---------------|-------------------|
| (1) | a. /tapu/ | → | [tappuri] | *[tampuri] | ‘a lot of’ |
| | b. /zabu/ | → | [zamburi] | *[zabburi] | ‘splashing sound’ |
| (2) | a. /hatu+tatu/ | → | [hattatsu] | *[hatsutatsu] | ‘development’ |
| | b. /hatu+gen/ | → | [hatsugen] | *[haggen] | ‘remarks’ |

However, gemination process observed in loanwords triggered the emergence of voiced geminates (McCawley 1968; Ito and Mester 1995, 1999 among others) in the phonology of Japanese. Especially, when a word-final sound is a voiced obstruent in a donor language, the sound is geminated with concomitant epenthesis, as in ‘dog’ being borrowed as [doggu] (Lovins 1973; Katayama 1998; Shirai 1999; Takagi and Mann 1994). Such a voiced geminate minimally contrasts with a voiceless geminate; there are minimal pairs like [kiddo] ‘kid’ and [kitto] ‘kit’ or [furakku] ‘Flack’ and [furaggu] ‘flag’ that show that this voicing contrast is indeed phonemic in the loanword phonology of Japanese. Yet, as mentioned above, voicing in geminates is aerodynamically challenging, and there is evidence that voicing in singletons and voicing in geminates is phonologically distinct. Kawahara (2003) shows that only voicing in geminates can be lost in reaction to the OCP(voi), which prohibits more than one voiced obstruent within a stem (also known as Lyman’s Law; see Ito and Mester 2003 and references cited therein). This is shown by the contrast illustrated in (3) and (4):

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(3) Voicing in geminates can be lost in response to the OCP(voi)¹

ge <u>bb</u> erusu	~	ge <u>pp</u> erusu	‘Göbbels (Proper Name)’
gu <u>dd</u> o	~	gu <u>tt</u> o	‘good’
be <u>dd</u> o	~	be <u>tt</u> o	‘bed’
do <u>re</u> ddo	~	do <u>re</u> ttō	‘dread’
do <u>gg</u> u	~	do <u>kk</u> u	‘dog’
ba <u>gg</u> u	~	ba <u>kk</u> u	‘bag’
do <u>ra</u> ggū	~	do <u>ra</u> kkū	‘drag’

(4) Voicing in singletons is not lost

<u>b</u> agii	‘buggy’	<u>b</u> ogii	‘bogy’
<u>b</u> obu	‘Bob’	<u>d</u> oguma	‘dogma’
<u>d</u> agu	‘Doug’	<u>d</u> aibu	‘dive’
<u>g</u> iga	‘giga’	<u>g</u> aburieru	‘Gabriel’
<u>g</u> ibu	‘give’	<u>g</u> aidansu	‘guidance’

To account for this asymmetry, following recent works on the P-Map hypothesis (Steriade 1997, 1999, 2001, 2004), Kawahara (2003) hypothesizes that voicing in geminates is more easily lost because voicing in geminates is harder to hear. Cross-linguistically, it is often the case that contrasts that are signaled by weaker cues are more prone to phonological neutralization (in addition to Steriade’s works cited above, Hura et al 1992; Jun in press; Kohler 1990; Ohala 1990; Ohala and Ohala 1993 among others). For example, preconsonantal consonants have many disadvantages in signaling place cues: they suffer from the lack of CV transition which provides the primary cues for place distinction, and they are often unreleased, which again weakens place cues (see Jun in press and references cited therein). As is well-known, preconsonantal consonants undergo place neutralization much more often than prevocalic consonants. The same generalization holds for the comparison between nasal and oral consonants in preconsonantal position. The vowel immediately followed by a nasal consonant is characteristically nasalized, and as a result VC transition cues are blurred due to anti-formants. As a consequence, preconsonantal nasals have weaker place cues than non-nasal consonants. Phonologically, nasals are more prone to undergo contrast neutralization (here place assimilation) than oral consonants in the same position.

Kawahara (2003) applies the same logic to explain the contrast between (3) and (4). Voicing is harder to detect in geminates, and therefore it is subject to phonological neutralization to a greater extent. More concretely, for example, the [atta]~[adda] contrast is less reliably perceived than the [ata]~[ada] contrast, so that it is not detrimental to lose voicing distinction in geminate pairs. In the words of Kohler (1990), it is an instance of “perceptual tolerated articulatory simplification” (see also Hura et al 1992; Huang 2001). Since voicing in geminates is harder to maintain, voiced geminates are phonetically undershot. As a consequence, relevant phonetic cues are weakened. This weakening of voicing cues results in lower perceptibility of voicing. Just as preconsonantal consonants are more likely to undergo place neutralization than prevocalic consonants, such

¹ Although this is an optional process, in a normal style of speech, it is more natural to use devoiced variants.

perceptually weakened voicing in geminates is more easily lost than more robustly cued voicing in singletons.

This paper reports on phonetic studies that aim to verify the hypothesis that voicing in geminates is less reliably perceived than voicing in singletons. As little is known about what kinds of cues are utilized to signal voicing in Japanese voiced geminates, I began with an acoustic experiment which was designed to figure out a set of acoustic cues that are significant in distinguishing voiceless and voiced consonants. The primary aim of this experiment was to see whether such cues are affected by singleton/geminate opposition, and if so, in what ways. In other words, the experiment was intended to investigate evidence from an acoustic side about whether voicing is harder to detect in geminates. The result of this experiment shows that some cues are indeed weakened in geminates, which might lead to the comparative difficulty of perceiving voicing in geminates. However, to complicate the story a bit, it is also found that some acoustic contrasts are enhanced in the context of geminates, which might facilitate perception of voicing in geminates. Yet, such patterns are subject to high inter-subject variability and the amount of enhancement is small, and therefore the conclusion of the first experiment is that it is doubtful that they compensate for the weakened cues sufficiently i.e., as a whole, voicing cues are indeed weakened in geminates.

With these observations in mind, the second experiment more directly tests the core hypothesis of this paper that voicing is harder to detect in geminates than in singletons. In order to most closely replicate the natural environments in which Japanese speakers hear voiced geminates, natural tokens recorded from the first experiment were used. In the experiment I asked Japanese speakers to identify the quality of voicing in a noisy environment. The result of this experiment clearly shows that voicing is hard to perceive in geminates while voicing in singletons is very accurately perceived. In summary, the two experiments reported here show the following points:

- (5) a) Some phonetic differences that are correlates of voicing are weakened in geminates.
- b) Some phonetic differences are enhanced next to geminates.
- c) Voicing in geminates is harder to perceive.

§2 reports the details of the acoustic experiment, and §3 reports the perceptual experiment. In the final section, I provide a general discussion of the results of these experiments.

2. Experiment I: Acoustics of voicing in geminates

As little work has been done on voicing in geminates in Japanese, the first experiment is designed to investigate the following three questions about voiced geminate in Japanese.

- (6) 1. What are the phonetic cues that signal voicing in Japanese?
- 2. How are such cues affected by singleton/geminate distinction?
- 3. If phonetic correlates of voicing are indeed affected by singletons/geminates opposition, does either position have disadvantages in signaling voicing contrast?

2.1. Methods

2.1.1. The speakers and recording

In this experiment, tokens of voiced geminates were recorded in a sound-attenuated booth.

Three native speakers of Japanese were recruited from the University of Massachusetts, Amherst community. They were all female and in their mid twenties. The dialects that they spoke were Shizuoka Japanese (Speaker E), Tokyo Japanese (Speaker T) and Hiroshima Japanese (Speaker W), although they were asked to speak the Standard Japanese (Tokyo Japanese) for the recording session. They were all paid for their time.

The pronounced target tokens were recorded to CD, digitized with a 22.050 KHz sampling rate and 16 bit quantization level, and submitted to acoustic measurements using Boersma and Weenink's (1992) Praat program.

2.1.2. The stimuli

The set of stimulus tokens consisted of 36 words, which were mostly nonce words.² To avoid letting the speakers know the intention of the experiment, 36 additional nonce words were added as fillers. The actual target words were all two-syllable words: the first consonant was fixed as [k], the second consonant was the target ([p], [t], [k], [pp], [tt], [kk], [b], [d], [g], [bb], [dd], [gg]) and three different vowels were used ([a], [e], [o]) for both the first syllable and the second syllable (henceforth V₁ and V₂, respectively). Some examples are thus: *kappa*, *kaba*, *kede*, *koddo*, *koto*, *kaka*, *kegge* etc. They were uniformly accented on the first syllable, i.e., had a HL contour, which is a default pattern in loanwords and nonce word pronunciation.

Each stimulus was written on a separate card in *katakana* orthography, which is conventionally used for loanwords. This type of orthography was used because voiced geminates are found only in loanwords. Six repetitions of each set were recorded with a short break in-between. The stimulus cards were shuffled between each session to randomize the order of the stimuli. In order to solicit natural utterances and avoid domain-edge strengthening effects (Fourgeron and Keating 1997; Cho and Keating 1997), the stimuli were embedded in the following frame sentence:

- (7) jyaa ____ de onegai
then, ____ with please
'please, (do something) with ____ (a casual register)'

The speakers were encouraged to produce sentences as if they had been talking to their friends. More specifically, they were asked to imagine a situation where they were preparing a party and they wanted their friend to fetch the things whose names were the target words. This is to avoid the hyper-articulation of the materials in an experimental environment, which might lead to conscious attempt to preserve closure voicing of geminates.

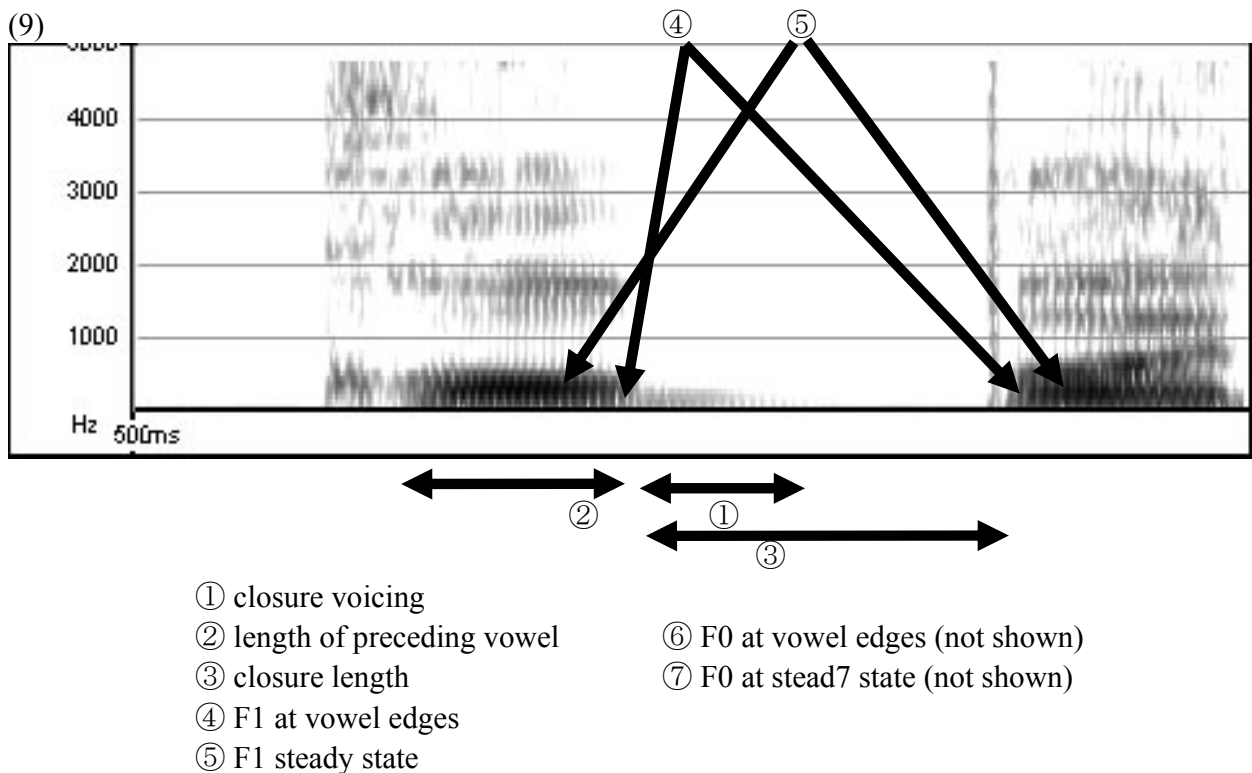
2.1.3. Measurement

All the measurements were done by using Boersma and Weenink's (1992) Praat program. Following the past literature that addresses reliable acoustic and perceptual correlates of the voiced/voiceless distinction (Lisker 1987; Kingston and Diehl 1994; Maddieson 1997; Raphael 1981; Stevens and Blumstein 1981), I measured the following values:

² It was impossible to completely exclude real words in this set; [kaka], [kaba], [kakka] are real words. Yet as they are all written in *katakana* orthography, at least [kaka] and [kakka], which are usually written in *hiragana*, should be hard to recognize as real words.

- (8)
- a) closure voicing
 - b) length of preceding vowel
 - c) closure duration
 - d) F0 of surrounding vowels
 - e) F1 of surrounding vowels

Closure voicing is an acoustic manifestation of glottal vibration during closure which appears as a voiced bar. It is expected that voiced consonants exhibit it while voiceless consonants do not. The second cue, the length of the immediately preceding vowel, is known to be longer before voiced consonants. The third correlate of voicing is closure duration; cross-linguistically, voiceless consonants have longer duration. Finally, F0 and F1 of the surrounding vowels are higher next to voiceless consonants. These measurement points (except for F0 which could not be shown in the spectrograms) are visually illustrated in the following sample spectrograms for [kobbo]]:



The more detailed explanation for how I measured these values are provided below when I discuss the results of these acoustic measurements.

2.2. Results

As the primary aim of this study is to see what the phonetic correlates of voicing are and how they are affected by singleton/geminate opposition, I did an ANOVA analysis in which consonantal length (2-level) and voicing (2-level) are repeated measures independent variables and subjects is as inter-subject variable.

The overall result shows that the phonemic difference in voicing is maintained both in singletons and geminates, and cued by the all measurement points listed above. The patterns are exactly what is expected in the light of cross-linguistic finding. However, some of these cues, most notably closure voicing, are weakened in geminates. At the same time all speakers compensate for this weakening in some way or another by enhancing a phonetic difference next to geminates or producing a phonetic difference not observed in singletons. I will discuss each of these aspects in more detail below.

2.2.1. Closure voicing

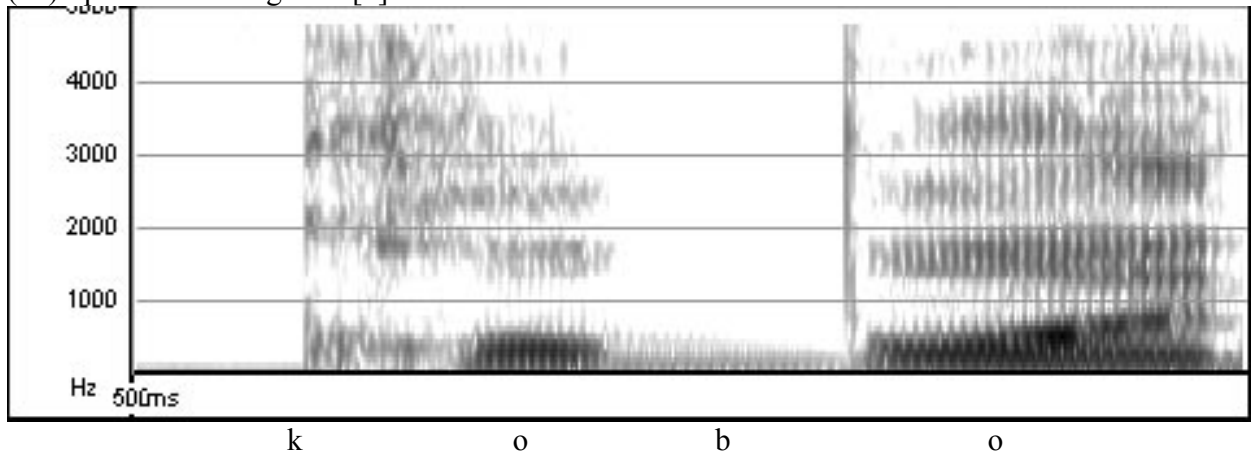
One of the most important voicing cues is duration of voicing within consonantal closure, acoustically realized as a so-called voice bar in low frequency range (Stevens and Blumstein 1981). The length of a voice bar was measured for each token, and the rate of closure voicing with respect to closure duration was calculated. The length of a voice bar was measured based on the presence of energy in low frequency range, as well as the presence of a periodic wave in the wave form. The onset of consonantal closure was in most cases acoustically unambiguous, signaled by sudden disappearance of vocalic formant energy. In case of gradual consonantal closure, which was sometimes observed for dorsal consonants, the disappearance of higher formants, especially F2 and F3, was used as a criterion for the onset of consonantal closure. The offset of consonant closure was set at its release, which was cued by burst noise. The values reported here do not include burst noise in closure duration.

The closure voicing percent with respect to the closure duration was calculated, and the result was fed into an ANOVA analysis in which voicing and length were two independent variables, and speaker is a between-speaker factor. The result is that voicing and consonantal length are both statistically significant: $F(1, 132)=775.462$, $p<.000$ and $F(1, 132)=1528.897$, $p<.000$, respectively. This shows that voicing distinction (naturally) affects the percent of closure voicing, and also that the length difference has an effect on the percent of closure voicing. This is because voiced geminates are frequently partially devoiced (see below for more). This is shown by the fact that length*voice interaction is also significant: $F(1, 132)=453.860$, $p<.000$: only geminates undergo partial devoicing and therefore the change in consonantal length necessarily affects the differences in voiced and voiceless consonants. There was no interaction observed between speaker and voicing or between speaker and length ($F(2,132)=2.453$, $p=.09$ and $F(2,132)=.142$, $p=.0868$). This indicates that the pattern is consistent throughout the speakers.³

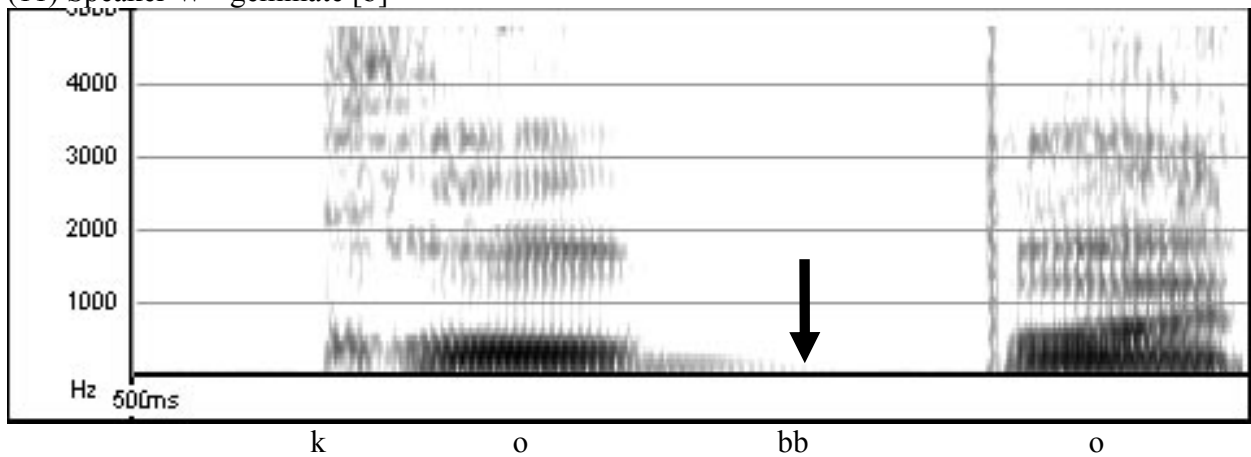
One of the most noticeable differences between singletons and geminates, which is clear from the statistical analysis above, is the fact that while almost all voiced singleton consonants maintain voicing throughout the closure, there are very few tokens of geminates in which such full voicing is observed. Some characteristic spectrograms are shown below. These show a minimal pair of voiced obstruents pronounced by Speaker W:

³ Place of articulation difference does not affect the percentage of closure voicing even for geminates ($F(2, 106)=2.883$, $p=.060$). This is somewhat surprising, as it is well-known that difficulty in voicing maintenance varies with the place of the obstruents due to the differences in size of the oral cavity and availability of active expansion (Hayes and Steriade in press; Kingston 2003; Ohala 1983; see§3.3).

(10) Speaker W - singleton [b].



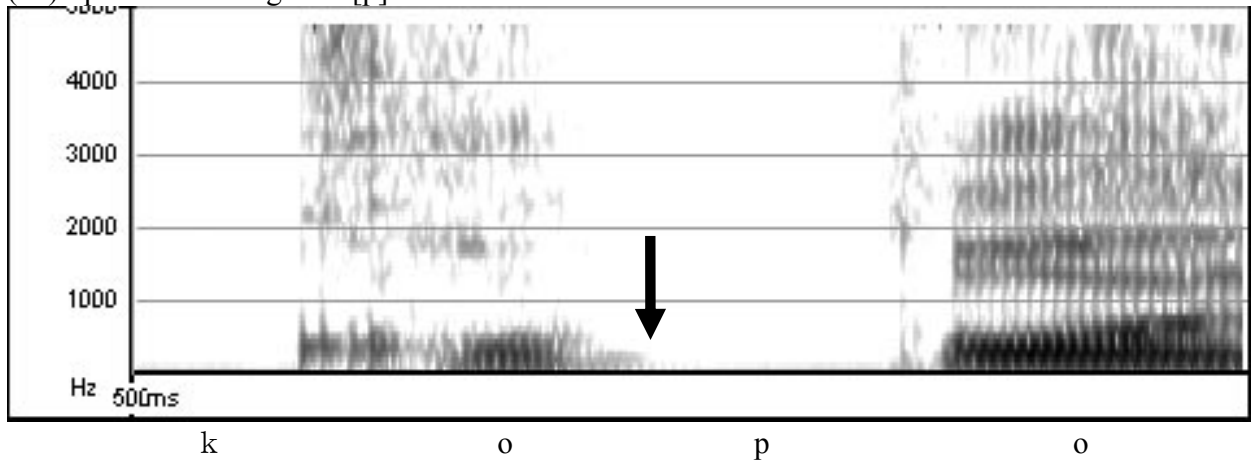
(11) Speaker W - geminate [bb]



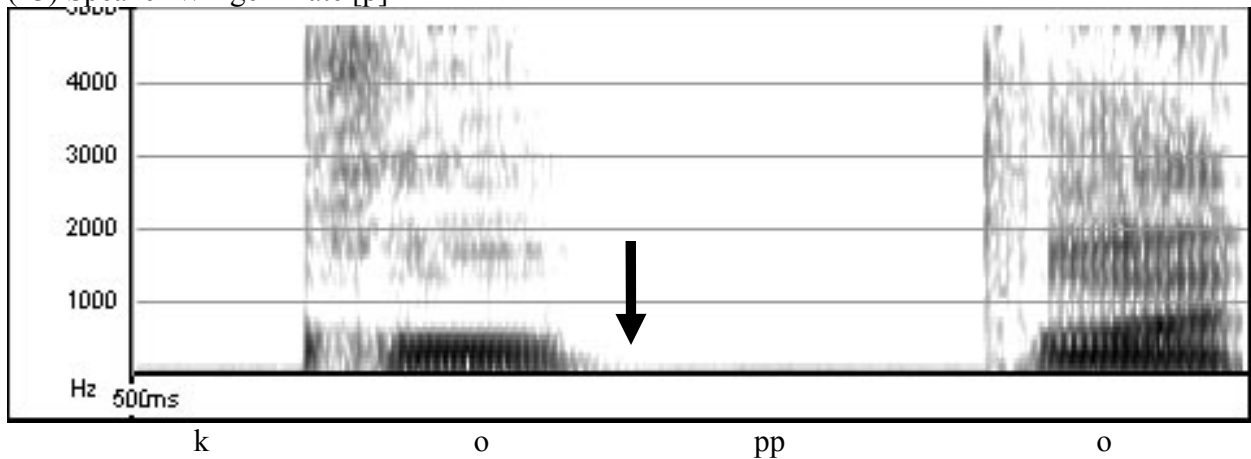
The first spectrogram is that of a singleton [b], and as seen, a voice bar continues throughout the closure. In contrast, in the second spectrogram, which has a geminate [bb], it is evident that voicing stops in an early phase of the consonantal long closure, indicated by an arrow.

This contrast between singleton and geminate voiced consonants is a very general pattern observed in all of the speakers; for example, Speaker T shows full voicing for all 54 tokens of voiced singletons while she exhibits no tokens of geminates in which voicing is maintained more than 80 percent of the closure. Speaker E and Speaker W show two instances of perfectly voiced geminates for [dd] and [bb], respectively, but all other tokens are at least partially devoiced. This is one of the reasons why voicing cues in geminates might be weakened. This point becomes more evident if we compare these spectrograms with those of voiceless consonants. Below are the spectrograms of [p] and [pp] from the same speaker:

(12) Speaker W - singleton [p]



(13) Speaker W - geminate [p]



As seen, even voiceless consonants have a little bit of voicing leakage due to the fact that glottal abduction and closure formation do not necessarily coincide. For singleton consonants, since the closure voicing is complete for voiced obstruents, the difference between voiceless and voiced consonants is still clear. However, in geminates, the difference in closure voicing between voiced and voiceless consonants become very small.

To see this attenuation of voicing difference in geminates, the average values of closure voicing percent for singletons and geminates are plotted in the graphs below. Here and throughout, the two first bars represent singleton values while the other two geminate values. Within each block, the first bar (solid) represents voiced consonants, the second (meshed) voiceless consonants:

(14)



As seen, for all speakers, singleton voiced consonants achieve almost 100 percent voicing maintenance; on the other hand, in geminates, even voiced ones have only around 30 to 40 percent of voicing maintenance rate on average (Speaker E=40.9%, Speaker T=38.5%, Speaker W=30.1%), suggesting that partial devoicing is prevalent. As evident from the graphs in (14), the differences between voiced and voiceless consonants are drastically reduced in geminates. Such weakening of closure voicing in geminates should have a strong impact on the perceptibility of voicing in geminates, as closure voicing is arguably a primary cue for voicing (Stevens and Blumstein 1981; see also Lisker 1987).

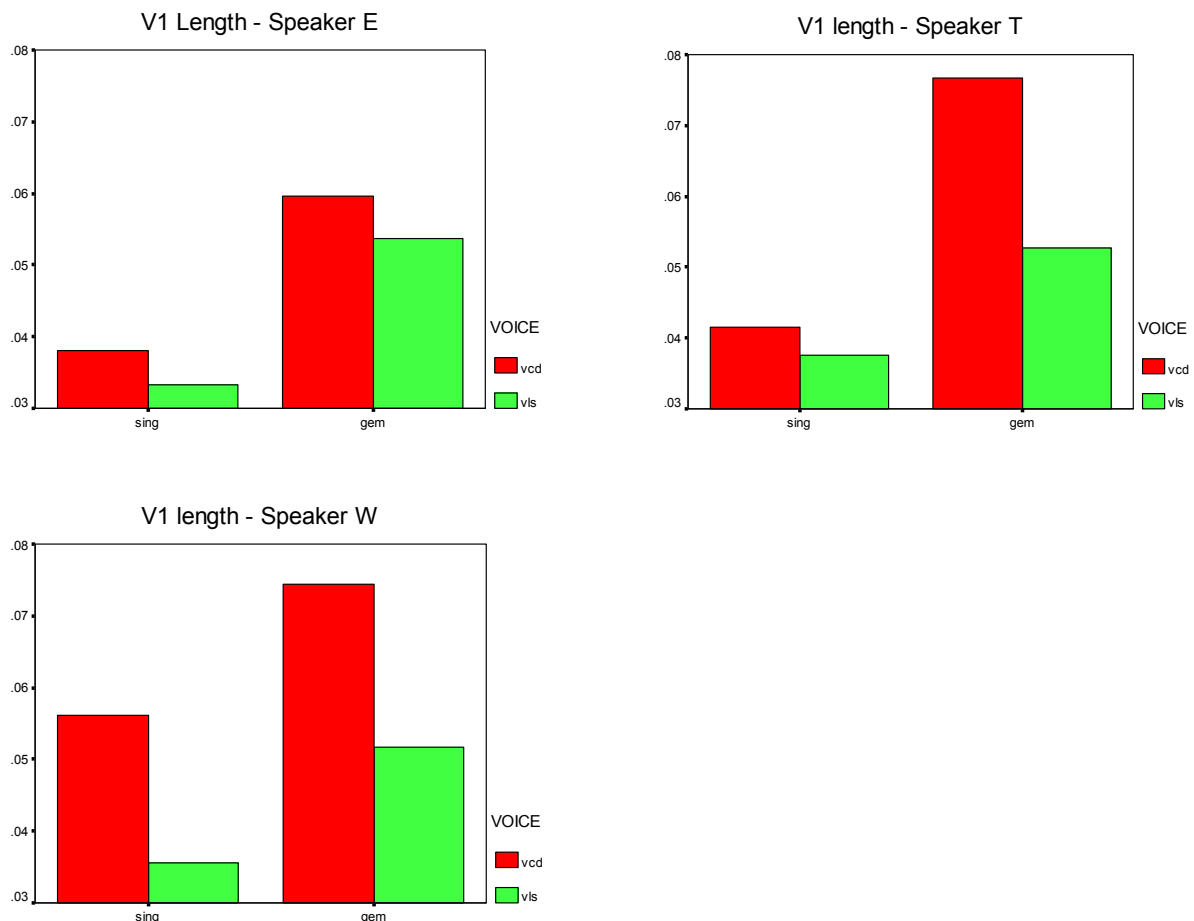
However, it is still important to note that even in geminates the contrast is maintained between voiced from voiceless ($F(1, 161)=273.631, p<.000$). Compared to voiced geminates that have around 30~40 percent of closure voicing, voiceless geminates have on average less than 10 percent of closure voicing. The average absolute duration of voicing in the geminates is around 40 milliseconds across all the speakers (Speaker E=42.2 msec, Speaker T=42.4 msec, Speaker W=38.3msec), which is not negligible. These values are significantly different from those for voiceless geminates, which are about 10 milliseconds (Speaker E=10.5 msec, Speaker T=9.0 msec, Speaker W=9.7 msec).

2.2.2. Duration of preceding vowels

The second phonetic difference that correlates with voicing distinction is the length of the immediately preceding vowel (V1). The onset of the V1 is set at the first periodic wave after aspiration of the preceding [k]. This is judged based on the beginning of a periodic wave in the wave form together with the appearance of clear F2 and F3 in the spectrogram. The offset is set at the onset of consonantal closure, which is signaled by the disappearance of F2 and F3.

An ANOVA analysis in which voicing and consonantal length are independent variables shows that voicing difference and consonantal length both have a statistically significant impact on the length of V1 ($F(1, 128)=515.837, p<.000$ and $F(1, 128)=277.847, p<.000$). This reflects the fact that V1 tends to be longer before voiced consonants as well as before geminates, as illustrated by the graphs below:

(15)



As cross-linguistically often observed, in the Japanese data at hand, the vowel is longer before voiced obstruents than voiceless ones (Denes 1955; Chen 1970; Raphael 1972, 1981; Castleman and Diehl 1995). As seen above, this is true for all the speakers before both singletons and geminates. It is interesting that this is true even for singletons in which V1 and the target consonants belong to a different syllable.

The fact that consonant length has an effect on V1 length captures the tendency to make preceding vowels systematically longer before geminates, contrary to the expectation that a vowel is shorter in a closed syllable (see Maddieson 1985 for a cross-linguistic overview; see also Smith 1995). One might thus suspect that a Japanese geminate does not close a preceding syllable, but this postulation is not tenable because geminates count as moraic, a characteristic of coda consonants (see, e.g., McCawley 1968; Poser 1990 for evidence). Even with such lengthening, however, the voicing contrast is still maintained in terms of V1 length before geminates ($F(1, 159)=269.191$, $p<.000$).

The interaction of voicing and consonantal length is also significant ($F(1, 128)=32.597$, $p<.000$), which shows that the extent of the difference in V1 length as a voicing cue varies whether it is before a geminate or a singleton. This is most clearly observed in Speaker T in which the difference is clearly enlarged in the context of geminates. A related observation is that this difference in V1 length exhibits high inter-subject variability. The interactions of the speaker variable with voicing and length are both statistically significant ($F(1, 128)=5.642$, $p=.004$ and $F(1, 128)=36.723$, $p<.000$, respectively); this shows that there is inter-speaker variation for the extent of V1 difference between before voiceless consonants and before voiced consonants, and how geminacy affects V1 length. In (15), it is observed that Speaker E has a relatively small difference between voiceless and voiced environments both for singletons and geminates. On the other hand, Speaker W shows relatively large differences in both environments. There is also inter-speaker variation in how geminacy affects V1 length. Of most interest, however, is the fact that the interaction of the all variables is significant ($F(1, 128)=18.642$, $p<.000$). This is due to Speaker T in which the V1 difference is quite enlarged before geminates (see (15)).

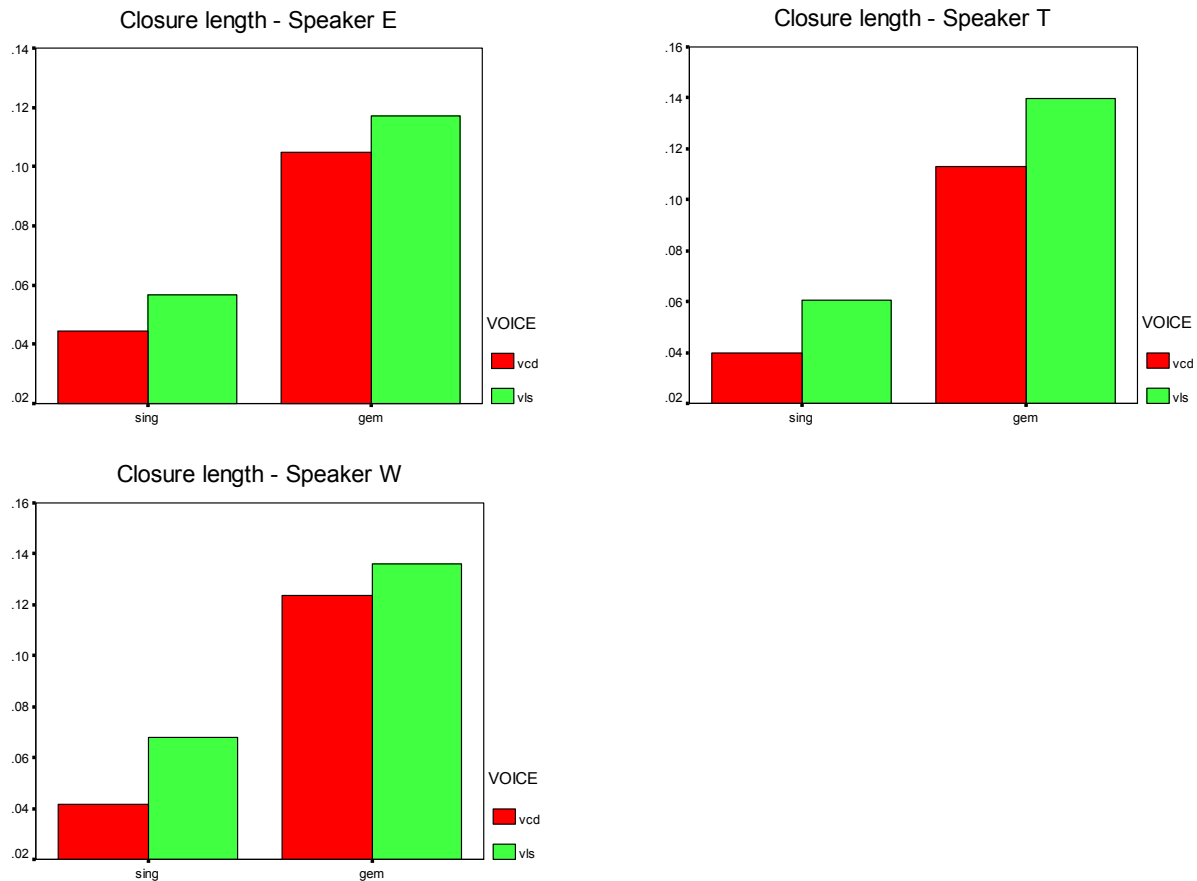
The fact that Speaker T has a larger difference before geminates than before singletons might be captured as an articulatory compensation effect. As geminates suffer from partial devoicing during closure, the speaker might attempt to enhance the contrast in V1 to signal voicing. In other words, to make up for the weakening of closure voicing as a voicing cue, the speaker enhances another cue. This is parallel with the case of non-prevocalic voicing in English: as VOT is unavailable as a cue to signal voicing in non-prevocalic positions, the contrast is expressed by the length of the preceding vowel; e.g., [be:d] and [bet]. That is, the loss of one phonetic correlate is compensated for by another emergent phonetic cue. What Speaker T does here is very similar to the English case in this respect. We observe below that a similar effect is exhibited by the other two speakers in different contexts.

2.2.3 Closure duration

The third observed difference between singletons and geminates for both singletons and geminates is closure duration. How consonant closure duration is measured is stated in §2.2.1. An ANOVA analysis suggests that, as cross-linguistically often observed (Westbury 1979; Ohala 1983: 195, 1990: 264), voiced consonants are shorter in duration, which presumably contributes to perception of voicing (Lisker 1957; Kingston and Diehl 1994). The voicing difference is statistically significant ($F(1, 127)=437.217$, $p<.000$); voiced consonants are systematically shorter. Length, quite naturally, exhibits a large significance; by definition, geminates have longer closure ($F(1, 127)=3199.789$, $p<.000$). The interaction of voicing and geminacy is not significant, which

suggests that the difference in closure length due to voiced/voiceless opposition is consistent regardless of singleton/geminate opposition. The graphs are shown below:

(16)



Although the closure duration difference to signal voicing is fairly constant between singletons and geminates, the percentage of voiced consonants with respect to voiceless consonants in terms of duration is of some interest. The following chart summarizes the rate of closure duration of voiceless consonants with respect to voiced ones for each place of articulation (the rows for singleton dorsal consonants are shaded because they are based on a very few tokens that did not spirantize):

(17) Speaker E

	singletons			geminates		
	vls	vcd	vcd/vls (%)	vls	vcd	vcd/vls (%)
lab	0.063	0.049	78	0.115	0.107	93
cor	0.059	0.043	72	0.120	0.107	90
dor	0.043	0.039	92	0.113	0.097	86

(18) Speaker T

	singletons			Geminates		
	vls	vcd	vcd/vls (%)	vls	vcd	vcd/vls (%)
lab	0.066	0.048	72	0.140	0.118	84
cor	0.052	0.030	58	0.137	0.107	78
dor	0.055	0.052	95	0.129	0.119	92

(19) Speaker W

	singletons			geminates		
	vls	vcd	vcd/vls (%)	vls	vcd	vcd/vls (%)
lab	0.077	0.050	64	0.145	0.125	86
cor	0.063	0.035	56	0.130	0.123	94
dor	0.060	0.039	64	0.123	0.115	94

What is evident is that the ratio of voiceless/voiced is much higher in geminates. This means that voiceless and voiced consonants are more similar to each other in terms of closure duration in geminates. Furthermore, in some cases (Speaker W's coronal and dorsal and Speaker E's labial), the ratio is above 90 percent, which means that voiceless and voiced consonants are almost identical in their length. This further implies that closure length, which is presumably one of the cues for voicing in Japanese, is attenuated (i.e., harder to detect) in geminates. This is yet another factor that might make voicing distinction in geminates harder to hear (see Ham 1998 for related discussion).

2.2.4 F0 at V2 onset

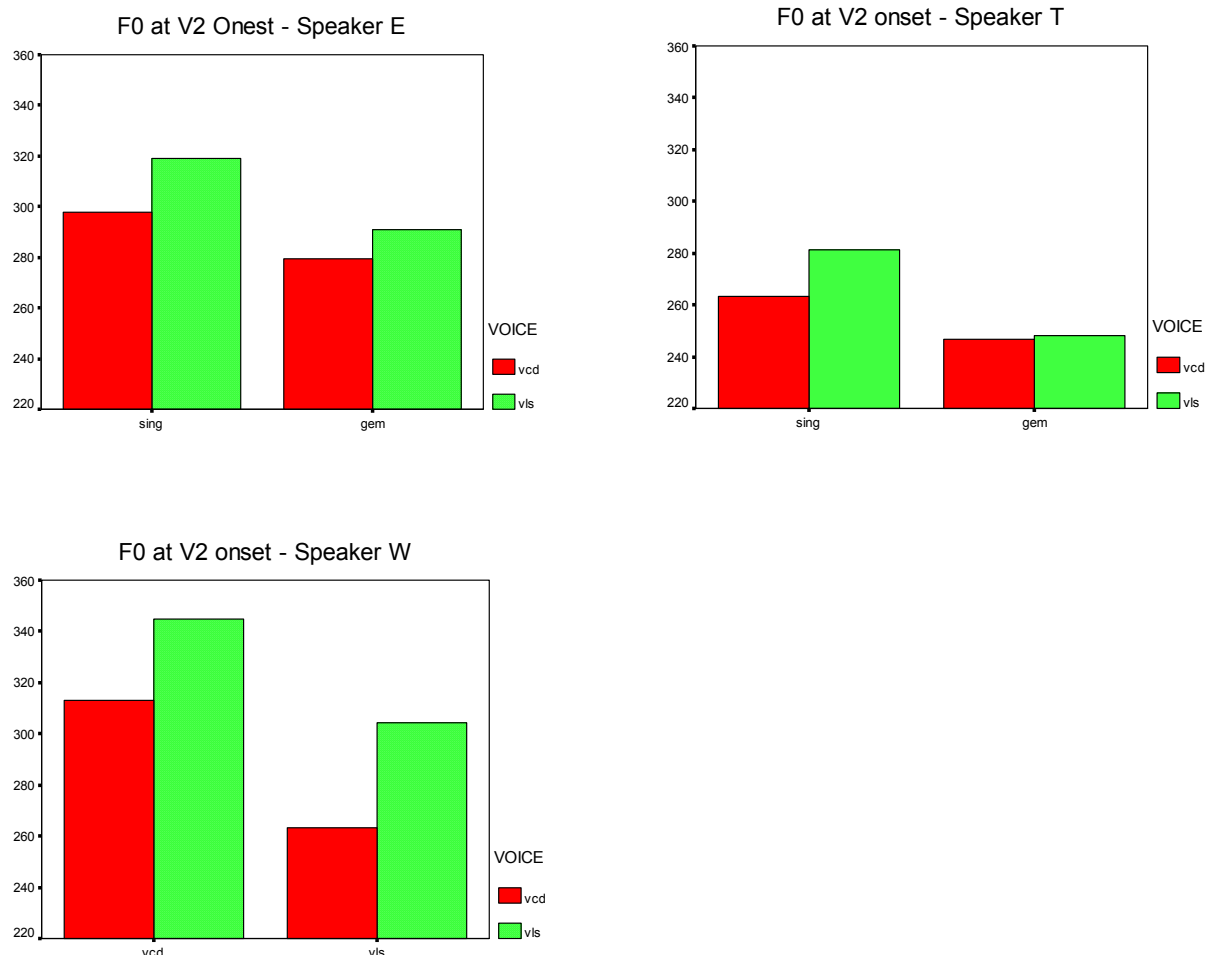
The fourth voicing cue is the value of F0 at V2 onset; the value of F0 at the first periodic wave right after the consonantal burst. The measurement was done by using Praat's detection function for pitch. Voicing threshold was lowered to 0.30, as sometimes the default setting did not detect the F0 value of the first periodic wave.

Cross-linguistically, it is observed that F0 is higher in vowels next to voiceless consonants (see Kingston and Diehl 1994 among others), and this is in general true for the Japanese speakers as well (the values of F0 at V1 offset are reported in §2.2.6). One exception is Speaker T for whom the contrast is almost lost after geminates. However, even though the difference is not high enough to be statistically significant, the tendency for F0 to be higher after voiceless consonants is still observed.

An ANOVA analysis shows that voicing and consonantal length both have a statistically significant influence on the F0 value of V2 onset ($F(1, 128)=174.173$, $p<.000$ and $F(1,128)=411.450$, $p<.000$). The interaction of these two factors is only marginally significant ($F(1, 128)=3.841$, $p=.052$); however, if we look at each speaker separately, the F0 differences between voiced and voiceless environments vary with singleton/geminate opposition, as seen in the graphs below. Speaker E and T have a smaller difference after geminates (and in fact Speaker T's difference is almost lost after geminates). Speaker W has a larger difference after geminates (around 32Hz for singletons and 40Hz for geminates). This observation is statistically supported by the fact

that the interaction of voicing, consonantal length and speaker is highly significant: $F(1, 128)=7.016$, $p=.001$).

(20)



Another observation about the F0 value at V2 onset: as seen in the diagrams above, the F0 values are systematically lower after geminates (recall that length has a statistically significant impact on the F0 values). There are two possible reasons for this. One is that since the tonal contour of the recorded tokens is HL, given long geminate closure, the fall in F0 is more drastic after geminates simply because there is more time to implement the HL fall (in a heavy syllable the fall starts at the first mora of the syllable (e.g., McCawley 1968:133-134)). The second possible explanation for this is that with geminate closure, the intraoral air pressure becomes so high that it is harder to keep transglottal air flow to vibrate the glottis, which results in lower F0 value (though cf. Kingston and Diehl 1994; see also some discussion below and in §2.2.5).

One simple experiment that would decide between these hypotheses would be to measure the same points using unaccented words, which have LH contour. If the first hypothesis is correct, then, since geminates afford more time to implement the rise, it is predicted that F0 at V2 onset is higher after geminates. If the second hypothesis is correct on the other hand, the effect of LH rise should be blurred because the high intraoral air pressure after geminates should prevent the pitch from rising so high.

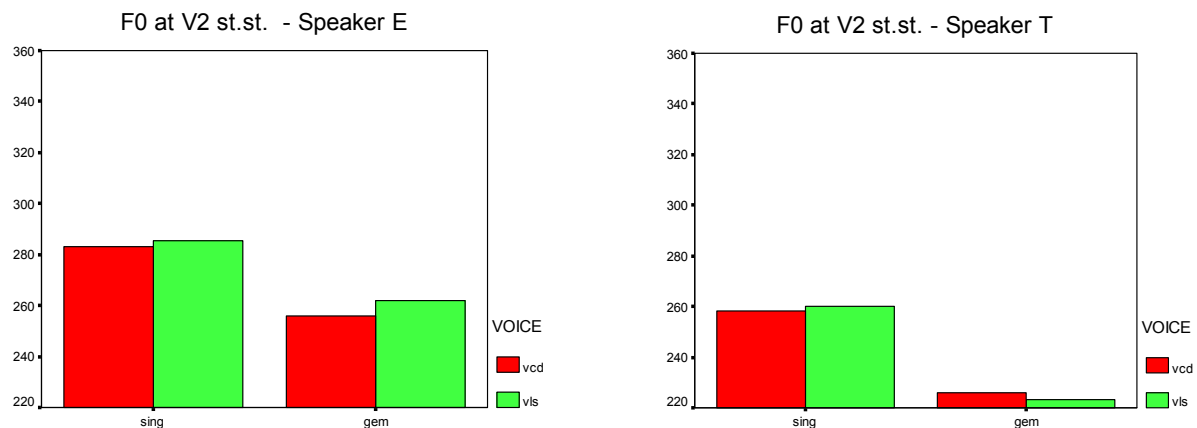
Another point that is worth stressing is the fact that Speaker W has a larger difference after geminates. This can be treated as a compensation effect in which the speaker attempts to enhance the voicing cue by F0 manipulation for geminates, whose closure voicing cue is weakened. Another related point is that, at least in Speakers E and W, the F0 difference is maintained after geminates, despite the fact that voicing usually stops during closure.⁴ These two points suggest that manipulation of F0 is not automatic but intentional (Kingston and Diehl 1994). If it were automatic, we would be unable to explain the fact that semi-devoiced voiced geminates have lower F0 in the following vowel. Also, the fact that a speaker can enhance the F0 difference after geminates suggests that it is possible to intentionally control the F0 value. Finally, to the extent that this manipulation is to enhance the voicing contrast, this is in line with Kingston and Diehl's view that such manipulation is essentially to enhance phonological contrasts.

2.2.5. F0 at V2 steady state

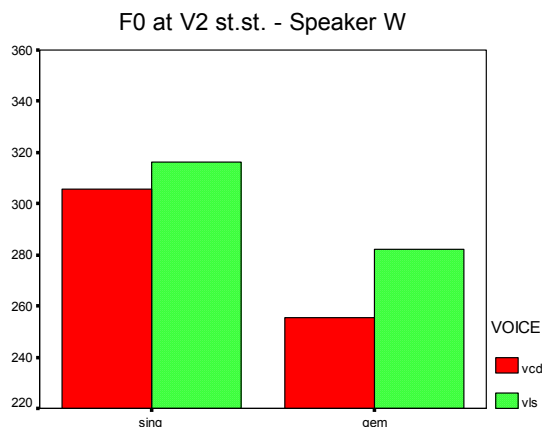
As seen above, F0 of V2 onset is systematically higher after voiceless consonants. F0 value of V2 steady state was also measured; the steady state was set at the sixth glottal pulse counting from the onset of the V2 (about 10 to 20 milliseconds away from the onset of the vowel).

Like F0 values at V2 onset, F0 is systematically lower after geminates ($F(1, 128)=864.700$, $p<.000$). The voicing difference also significantly affects the F0 value ($F(1, 128)=28.201$, $p<.000$), although this is only true for Speakers E and W, as discussed more fully below. The interaction of length and voicing is also significant ($F(1, 128)=5.762$, $p=.018$), as both Speaker E and W have larger differences after geminates.

(21)



⁴ A similar fact is reported in English [+voi] consonants where F0 depression is observed next to [+voi] regardless of the presence of the actual voicing during the closure (Kingston and Diehl 1994 citing an unpublished work by Caisse (1982)).



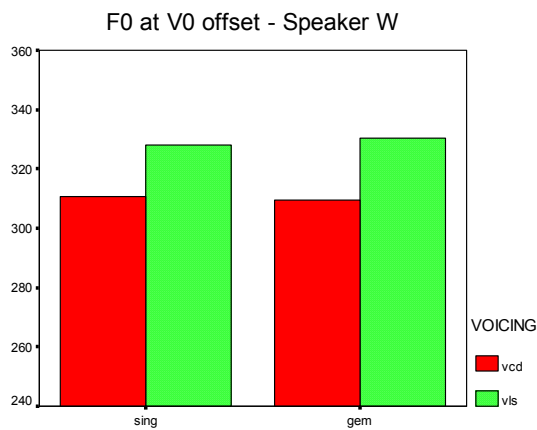
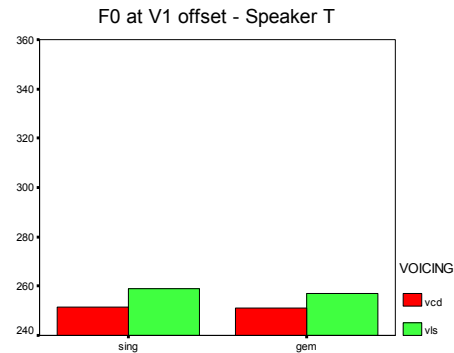
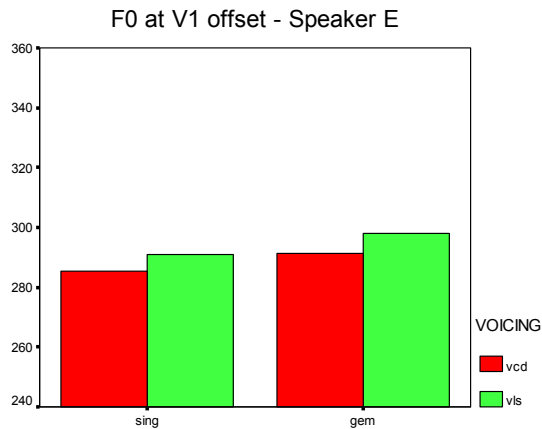
Speaker T does not show any difference between post-voiceless position and post-voiced position ($F(1, 34)=8.705$, $p=.787$): in fact, after geminates, F0 is higher after voiced geminates, contrary to the expectation. More interesting is Speaker E, for whom the difference after singletons is not statistically significant, $F(1,49)=.655$, $p=.422$, (although the general tendency that F0 is higher after voiceless consonants is maintained), while the difference is significant after geminates ($F(1, 51)=8.665$, $p=.005$). This pattern observed in Speaker E - that the F0 contrast emerges only after geminates - can again be treated as a compensation effect. Voicing is weakened in geminates, so that the speaker might attempt to signal a voicing contrast in a way that is specific to geminates so that voicing contrast is not lost. Similarly, Speaker W has a larger difference after geminates, which can also be captured as a compensation effect. The reason why Speaker W has relatively large compensation (the difference in singletons is about 11Hz while the difference in geminates is about 27Hz), compared to Speaker E, might be because she maintains closure voicing least among the three speakers (Speaker E=40.9%, Speaker T=38.5%, Speaker W=30.1%; see §2.2.1). Reflecting such inter-speaker variability, the interaction of voicing and length among three speakers is highly significant ($F(2, 128)=1128.675$, $p<.000$).

The behavior observed in Speakers E and W again provide support for Kingston and Diehl's (1994) claim that controlling F0 next to voiced/voiceless consonants can be deliberate. This is because speakers can manipulate the difference in F0 depending on the context, i.e., depending on whether the vowel is next to a singleton or a geminate.

2.2.6 F0 at V1 offset

The voicing contrast is also cued by the F0 value of the preceding vowel. F0 is higher before voiceless consonants: An ANOVA analysis shows that the influence of voicing on F0 at V1 offset is very significant: $F(1, 128)= 82.806$, $p<.000$. On the other hand, unlike the case of F0 at V2, no systematic effect of singleton/geminate opposition on F0 value is observed: $F(1,128)=3.078$, $p=0.82$. As seen in the graphs in (22), although Speaker E has a slightly higher F0 value before geminates, Speakers T and W do not show such difference (reflecting this, the interaction of consonantal length and speaker is significant: $F(1, 128)=4.721$, $p=.011$). The interaction of voicing and speaker is also significant ($F(1, 128)=13.280$, $p<.000$); Speaker W has a much larger difference compared to Speakers E and T.

(22)

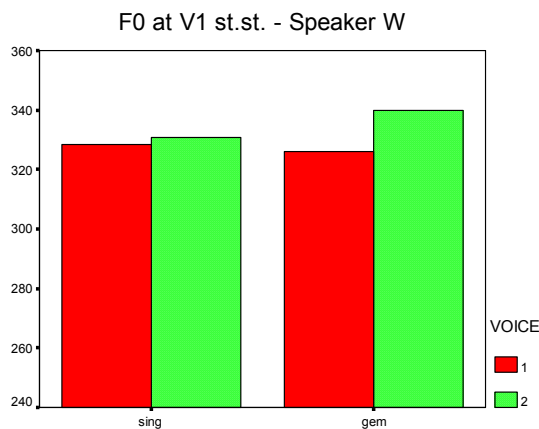
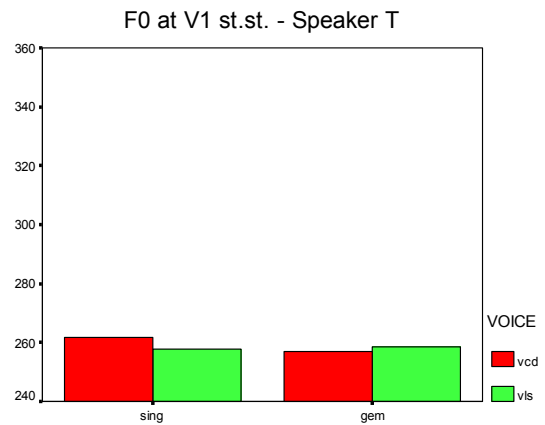
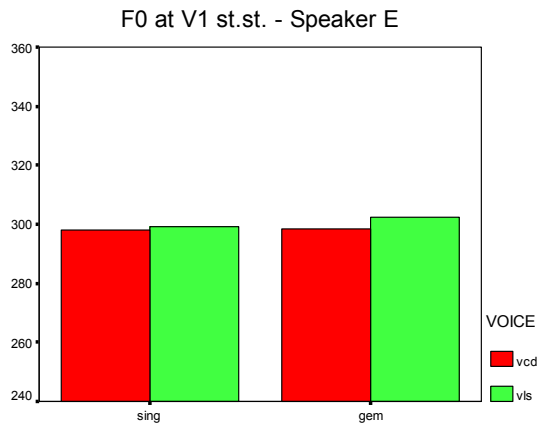


2.2.7. F0 at V1 steady state

The F0 values at V1 steady state were also measured. The measurement point was basically set at the sixth glottal pulse away from the offset of the vowel (about 10 to 20 milliseconds away from the consonant closure). There are some cases before voiceless consonants in which V1 is very short so that the sixth pulse is located too close to the transitional state of the first consonant [k]. In such cases, the midpoint of the vowel was calculated and F0 of that point was measured.

The pattern of F0 at V1 steady state is quite similar to the pattern of F0 at V2 steady state. Speaker E has no difference before singletons ($F(1, 48)=.126, p=.724$), but shows a difference before geminates ($F(1, 52)=5.731, p=.020$). Speaker W exhibits larger differences before geminates, while Speaker T does not show any difference either before singletons or geminates. Overall, the differences between pre-voiceless and pre-voiced positions are small albeit statistically significant: $F(1, 125)=4.826, p=.01$. The length has no effect on the F0 value at the steady state, just as no effect was observed at the offset point, $F(1, 125)=0.87, p=.353$. This is the tendency observed throughout the speakers, hence no interaction between speaker and length ($F(2, 125)<1$).

(23)

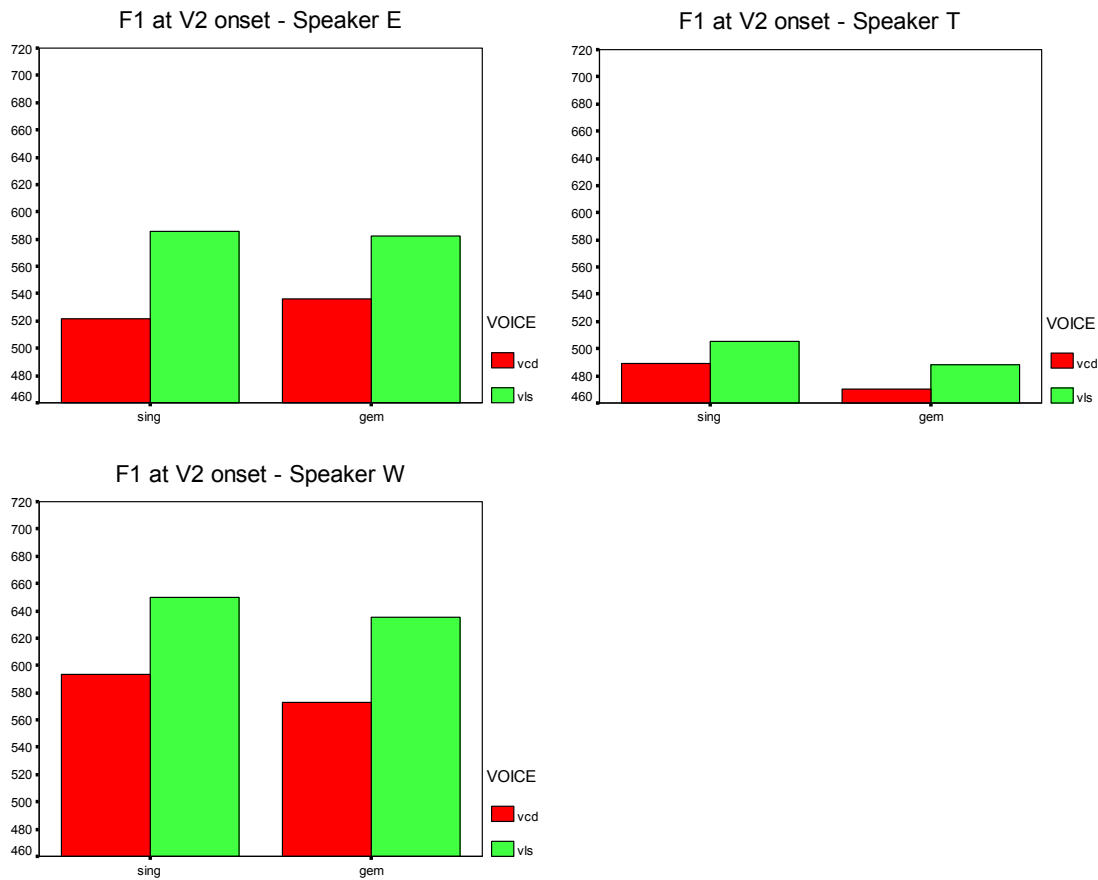


2.2.8 F1 at V2 onset

Cross-linguistically, the F1 value is also known to be higher next to voiceless consonants (Benki 2001 for an overview). To see if such tendency is also observed in Japanese, F1 of both at the V2 onset and V2 steady state was measured, using the formant detection function of Praat. The onset was set at the first periodic wave after the burst, and the steady state is at the sixth glottal pulse.

An ANOVA analysis with the two independent variables (voice and length) shows that voicing affects F1 of V2 onset $F(1, 127)=113.988, p<.000$. As expected, for all the speakers, F1 values of V2 offset are higher after voiceless consonants. The length also has an impact on the F1 value ($F(1, 127)=6.099, p=.015$): for Speakers T and W, F1 is systematically lower after geminates. The F1 differences due to voicing distinction are similar in post-singleton and post-geminate positions: hence, no interaction of length and voicing is observed ($F(1, 127)=.295, p=.588$).

(24)

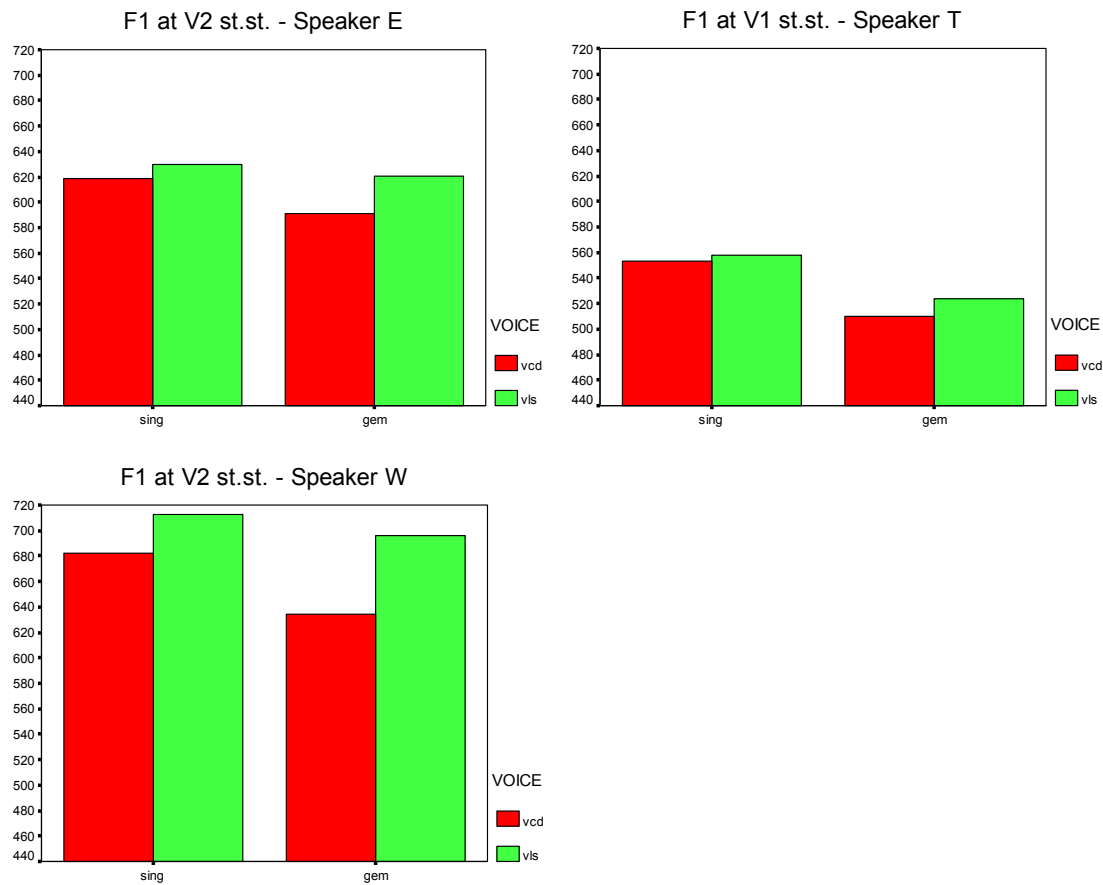


Here again, as was the case in F0 values at V2 onset, the phonological distinction between voiceless and voiced consonants has a significant effect on the F1 value at this position, despite the fact that voicing itself stops during closure, and hence at the CV juncture, the voiced geminates are phonetically voiceless. This suggests that the difference of F1 appearing next to voiceless/voiced consonants is not an automatic effect, but instead speakers can intentionally manipulate its values.

2.2.9. F1 at V2 steady state

F1 values at V1 steady state exhibit very consistent patterns across the speakers: (i) F1 is higher after voiceless consonants, (ii) the difference is larger after geminates and (iii) F1 is slightly lower after geminates than after singletons. Voicing has an effect on F1 at V2 steady state, $F(1, 126)=71.162, p<.000$. Geminates also lower F1: length exhibits a statistically significant difference ($F(1, 126)=99.573, p<.000$). As observed in (25), the differences of F1 due to a voicing distinction are enlarged after geminates; therefore, the interaction of voicing with length is significant, $F(1, 126)= 10.580, p=.001$.

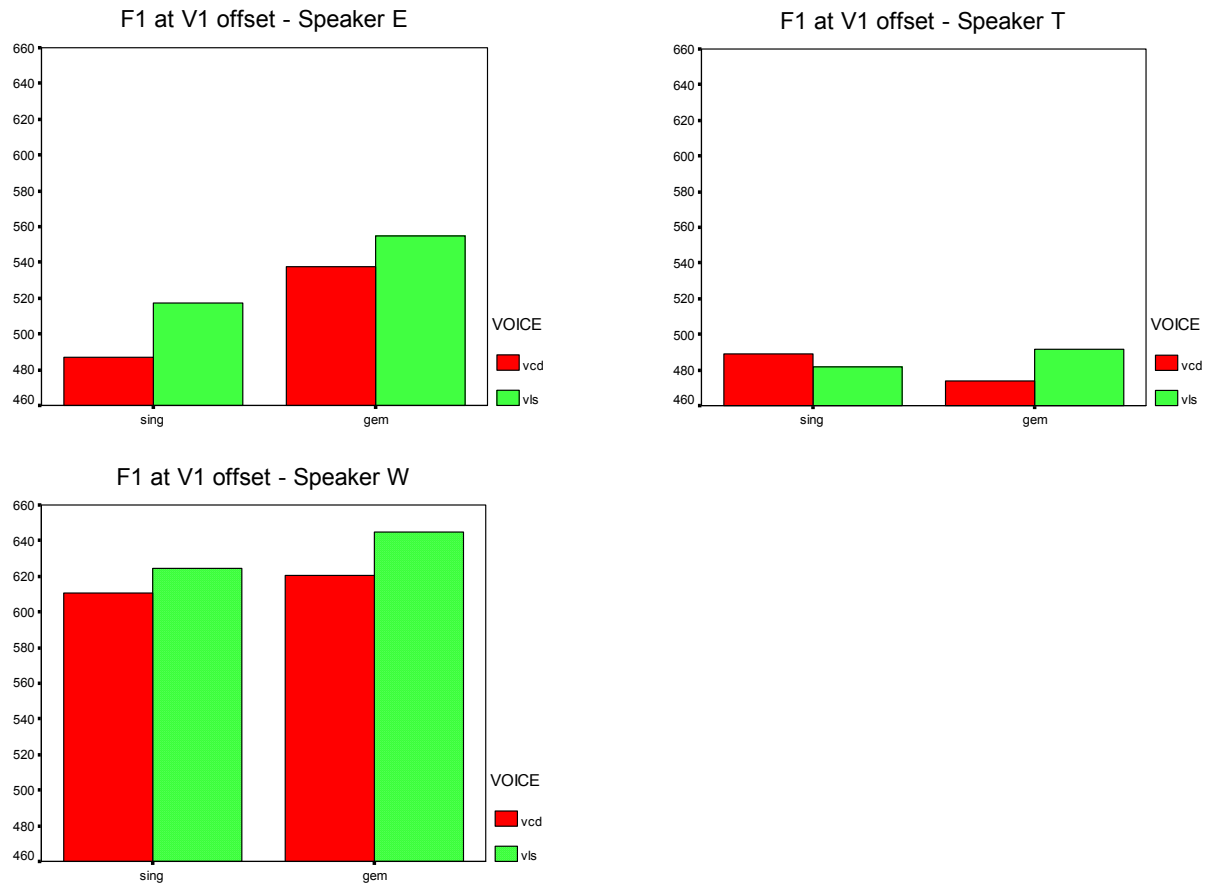
(25)



2.2.10. F1 at V1

As seen above, the F1 value is in general higher next to voiceless consonants. This tendency is true at the F1 of V1 offset, except for Speaker T for whom the difference between voiced and voiceless is neutralized before singletons ($F(1, 126)=8.756, p=.004$). Singleton/geminate contrast also has a significant impact on the value of F1: the F1 is systematically higher before geminates, $F(1, 126)=9.599, p=.002$. Neither the interaction of voicing and length nor the interaction of all three factors were significant ($F(1, 126)=.586, p=.445$ and $F(2, 126)=1.360, p=.360$).

(26)



I cannot think of a good reason why F1 is higher before geminates in V1 offset than in singletons. One very tentative hypothesis is that geminates might involve slower (and hence more gradual) closure so that F1 lowering due to consonantal constriction is less drastic in geminates.

The F1 value at the V1 steady state does not show any statistical significance ($F(1, 123)=0.01$, $p=.997$ for voicing difference). This is true for both singletons ($F(1, 123)=.050$, $p=.823$) and geminates ($F(1, 159)=0.35$, $p=.852$).

2.4. Discussion

The purpose of this acoustic experiment was to see what kinds of cues are used to signal voicing in Japanese, and how these cues are affected by singleton and geminate distinctions. As shown in the preceding discussion, Japanese seems to utilize various kinds of voicing cues that are known to signal voicing cross-linguistically. We have also seen that some phonetic differences are weakened or attenuated in geminates, while some speakers exhibit some phonetic differences only in the environment of geminates or make some phonetic differences more prominent surrounding geminates. Both effects can be captured as a contrast compensation effect. The overall result is summarized below:

(27)	<u>Phonetic cues</u>		<u>Change in geminates</u>
	a. closure voicing	→	Highly weakened in geminates.
	b. V1 length	→	Difference enlarged in geminates for Speaker T but not others.
	c. closure length	→	The ratio of vcd/vls is closer to 1 in geminates.
	d. F0 at V2 onset	→	A larger difference after geminates in Speaker W; A smaller difference after geminates in Speaker E and T.
	e. F0 at V2 st. state	→	A difference appears only after geminates in Speaker E; A larger difference after geminates in Speaker W.
	f. F0 at V1 offset	→	None.
	g. F0 at V1 st. state	→	A difference appears only after geminates in Speaker E; A larger difference before geminates in Speaker W.
	h. F1 at V2 onset	→	Speakers T and W have a larger difference before geminates.
	i. F1 at V2 st.state	→	A difference is enlarged after geminates.
	j. F1 at V1 offset	→	None.

One generalization that holds for all the speakers is that closure voicing and closure duration are attenuated in geminates, but F0 and F1 differences are enhanced in some dimension or another surrounding geminates (modulo F0 at V2 onset for Speaker E and T). Therefore, whether voicing in geminates is harder to perceive than voicing in singletons depends upon which set of cues is more important for voicing perception in Japanese speakers. To address this question, a series of experiments is necessary to see which cues contribute to voicing perception, and to what extent. This task, however, is left for future research.

However, it seems reasonable to speculate that overall voicing cues are weakened in geminates. First, internal voicing, which arguably constitutes a primary cue for voicing perception, is highly weakened in geminates (Stevens and Blumstein 1981). Second, it should be noted that such compensation effects are subject to inter-speaker variation. In fact, none of the strategies is taken by all three speakers, except for an F1 difference enhancement at V2 steady state. For example, although the F0 difference at V2 onset is enhanced after geminates for Speaker W, the opposite pattern holds for Speakers E and T. Thus, unless such cues are integrated in some way (Stevens and Blumstein 1981; Kingston 2004) so that such integrated cues are consistently enhanced in the context of geminates, it is doubtful that such enhancements provide a reliable perceptual cue. Even if such enhancements are indeed intended to compensate for the weakening of other cues, it is also doubtful that the degree of compensation is enough. For example, Speaker E's F0 at V2 steady state exhibits a difference only after geminates, but the difference that emerges is around 6 Hertz. Speaker W's enhancement of F0 differences after geminates is only 8-10Hz. It seems unlikely that such small differences have a significant perceptual effect. In sum, in contrast to the rather strong and systematic weakening of closure voicing, compensations are subject to inter-speaker variability, and the effects seem very small. So from an acoustic point of view, it seems likely that voicing cues are weakened in geminates.

This conclusion is supported by the result of the perceptual experiment, reported below in §3. In Experiment II, the result indeed shows that compensation is not enough; that is, the voicing contrast in geminates is harder to perceive even for Japanese speakers.

Before closing this section, even though it is not directly related to the main purpose of this experiment, it is worth noting that we find two pieces of evidence that F0 or F1 differences in

vowels next to voiced/voiceless consonants are not derived automatically, but due to speaker's intentional manipulation. First, even though phonologically voiced geminates are usually devoiced at the time of release, a difference is still observed in F0 and F1 at the following vowel. Second, F0 and F1 differences are sometimes enlarged in the context of geminates, suggesting that such manipulation is intentional, to maintain a phonological contrast.

3. Experiment II: perceptual experiment

In order to more directly test the hypothesis that voicing in geminates is harder to hear, a perceptual experiment was conducted. The primary aim of this experiment was to see how Japanese speakers perceive voicing in singletons and geminates in natural environments. In order to replicate this situation most accurately, therefore, natural tokens recorded in the first experiment were used. However, if we use natural tokens and nothing else, Japanese speakers might judge voicing in singletons and geminates almost all the time, as voicing is phonemic in loanword phonology. To overcome this problem, the stimuli were covered by cocktail party noise so as to raise the confusion rate.

With the observations obtained from the first experiment - that acoustic cues for voicing in geminates are overall attenuated - the prediction is that voicing in geminates is perceived relatively poorly compared to voicing in singletons. The result below shows that it is indeed the case.

3.1. Methods

3.1.1. Stimuli

From the pool of data obtained in Experiment I, for each speaker, one representative example of each token was chosen. The total number of the stimuli was therefore 108 (3 speakers $\times$ 3 vowels $\times$ 3 places of articulation $\times$ 2 consonantal lengths $\times$ 2 voicing types). Tokens that showed phonetic distortion or spirantization were excluded as much as possible (for the case of singleton [g]s, tokens with least spirantization were chosen). For voiced geminates, among the tokens that had no distortions, those that had closure voicing duration that was closest to the average value were chosen in order to use a representative token of natural voiced geminates.

Cocktail party noise was recorded at the Department of Linguistics of University of Massachusetts, Amherst at a post-colloquium party, using a DAT recorder. The recorded sound was transferred to a PC, and divided into 3-second noise stretches by Praat. Parts that have clear laughter or utterance were avoided. To produce the noise files that were used in the experiment, six files of such 3-second stretches were randomly chosen and superimposed on top of one another. Twelve kinds of such noise files were created.

To equalize the amplitude of all the stimuli, the maximum amplitude of all the stimuli was adjusted to 0.50 Pascal⁵ by using the modify function of Praat; the amplitude of the noise files was modified to 0.45 Pascal in the same way. One noise file was randomly chosen and was superimposed on each stimulus word. After the stimuli files and the noise files were combined, the

⁵ Pascal is a unit of pressure. The more commonplace unit for dB for x can be calculated as $10 \times \log_{10} (x/2 \times 10^{-5} \text{Pa})$.

edges of the combined files where only noise was present were trimmed off. After this process, each stimulus file was approximately 1.5 second long.

Some combinations of the sound file and the noise file turned out to yield some undesirable effects, most notably distortion of place features, which were reported by people who participated in the pilot experiment. In such cases, I assigned a different noise file to avoid this problem.

3.1.2. Subjects

In the main experiment, seventeen native speakers of Japanese were recruited. The speakers that participated in the first experiment were excluded as they might have some advantage when hearing their own voice. The participants were mainly recruited from the University of Massachusetts, Amherst community. All the subjects had normal hearing and were free of any speech disorders. Some of them had a basic knowledge about linguistics, but none had had extensive phonetic hearing training. The range of dialects that the speakers spoke was diverse, including Chiba Japanese, Tokyo Japanese, Shizuoka Japanese, Ibaragi Japanese, and Osaka Japanese. Two of the subjects were bilingual speakers of Japanese and English, but the inclusion of their data did not seriously change the overall result; hence they are included in the result reported below. All the subjects were paid or given extra credit if they were taking a linguistics class.

3.1.3. Task

The subjects were asked to sit in front of a computer and listen to the stimuli over headphones. During a practice session, they were instructed to adjust the volume to a comfortable listening level. In main sessions, as soon as they heard a stimulus, two choices written in *katakana* orthography minimally different in terms of voicing showed up in a screen. For example, for the stimulus [kappa], the two choices were ‘kappa’ and ‘kabba’ written in *katakana*. This use of *katakana* should bias the speakers that stimuli are loanwords, in which voiced geminates are allowed. The task was to push the button based on what they heard. One stimulus, as mentioned above, lasted for about 1.5 seconds. In order to make sure that speakers respond to all the stimuli, there were no time limits assigned to respond, although all the speakers responded within a second at most. The listeners were not given feedback about the correctness of their response.

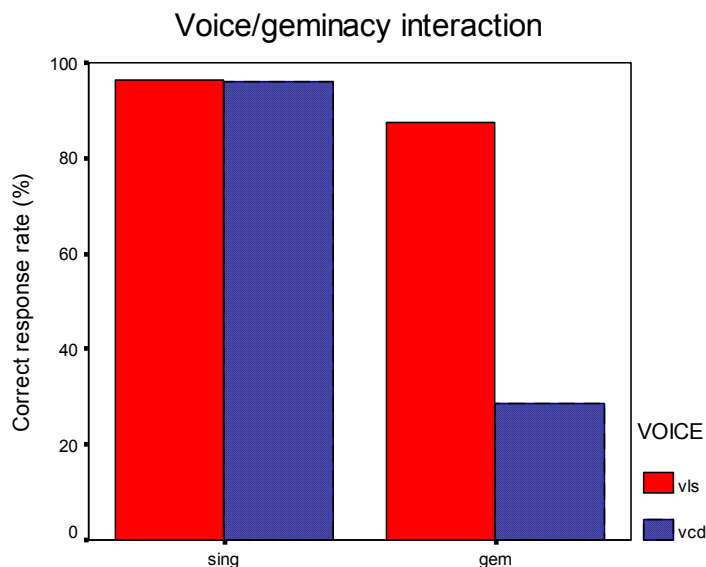
One session consisted of three blocks; each block contained all the types of stimuli pronounced by one speaker. One block contained 36 types of tokens, and therefore one session contained 108 stimuli as a total. One session usually ended in a few minutes. The entire experiment consisted of 8 such sessions.

3.2. Results

The result of this experiment clearly shows that voicing is indeed harder to hear in geminates. First, listeners' accuracy (i.e., the proportion of correct answers) for each item was calculated across all eight repetitions. Then, an ANOVA was applied to the data in which consonant voicing (2-levels), consonantal length (2-levels), and consonant place (3-levels) were independent variables. To simplify the analysis, the two other factors (speaker and vowel quality) were averaged over. Vowel quality difference is not quite significant ($F(2,32)=3.028$, $p=.062$). Although the speaker variable exhibits a statistical difference, the mean values are not so different (Speaker E=78.5%, Speaker T=76.2, Speaker W=76.7%).

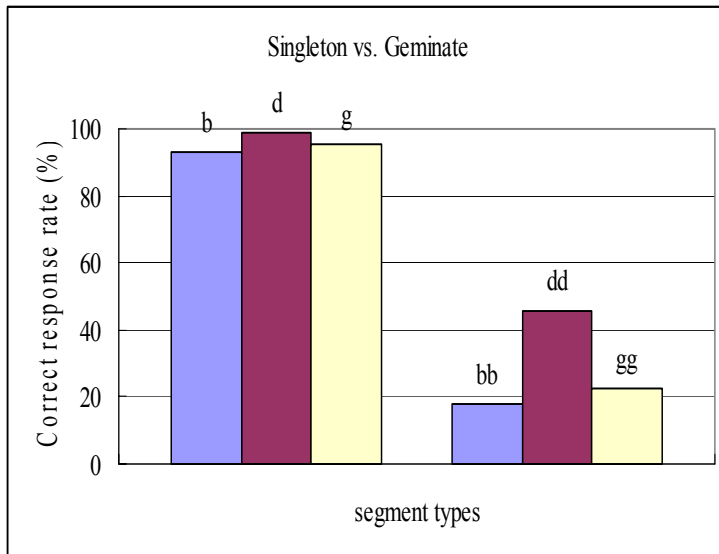
Most importantly, there is a very large, statistically significant difference in the response rate between singletons and geminates: $F(1, 16)=980.955$, $p < .000$, and the interaction of voicing and consonantal length is also highly significant: $F(1,16)=45.437$, $p < .000$. The graph in (28) illustrates the general interaction of geminacy with voicing.

(28)



As illustrated in the first two bars in (28), when the target is a singleton consonant, both voiceless and voiced consonants are judged correctly more than 95 percent ($vls=96.4\%$; $vcd=96.0\%$), and the difference is not at all significant ($F(1,16)=.746$). Voiceless geminates are judged nearly as well (87.6%). On the other hand, voiced geminates suffer from significant rate of misjudgment: the correct judgment rate goes down to 28.7%. This shows that voicing is indeed hard to detect in geminates, while voiced singleton consonants do not suffer from such a problem.

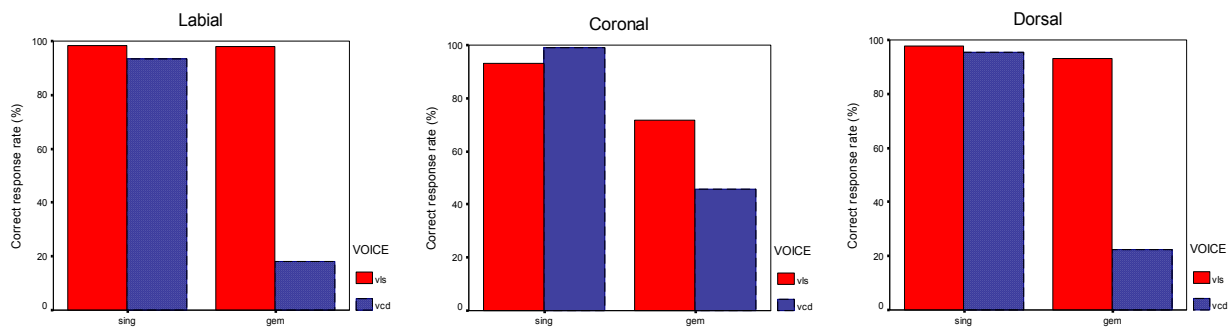
(29)



The graph in (29) also illustrates this point, but this time for each place of articulation; singleton [b], [d], [g] are judged almost 100 percent ([b]=93.3%, [d]=99.1%, [g]=95.3%), while geminate [bb], [dd], [gg] are judged quite poorly ([bb]=17.9%, [dd]=46.6% and [gg]=22.3%). When we compare three places of articulation, both in geminates and singletons, the coronal voiced obstruents show the best performance, compared to labials and dorsals.

For each place of articulation, the tendency of voiced geminated to be judged inaccurately basically holds. The graphs below show the interaction of geminacy with voicing for each place of articulation.

(30)



In general, voiced consonants are misjudged when they are geminated, while all the other types of consonants are judged accurately almost all the time.

Interesting differences are observed between coronals on the one hand and labials and dorsals on the other. In singletons, in labials and dorsals, voiceless consonants show better performance, although the pattern is reversed in coronals. Moreover, in labials and dorsals, voiceless geminates (as well as singletons) are judged correctly almost 100 percent of the time

([pp]=98.0%, [kk]=93.0%) while for coronals, some geminate [t]s are misheard as voiced (28%). In addition, the correct response rate for [dd] is much higher than the correct response for [bb] and [gg] ([bb]=17.9%, [dd]=46.6% and [gg]=22.3%; see (28)). Indeed, the voice/place interaction is statistically significant between coronals on the one hand and labials and dorsals on the other⁶:

- (31) Coronal vs Labial: $F(1, 16)=37.067, p<.000$
Coronal vs Dorsal: $F(1, 16)=74.897, p<.000$

3.3. Discussion

The perceptual test supports the hypothesis that voicing is harder to hear in voiced geminates. This suggests that the weakened cues - internal voicing and closure length - are not compensated enough by manipulation of F0 and F1 surrounding geminates (see §2). We can conclude from this that closure voicing and/or closure length are important cues to signal voicing in Japanese. Exactly how each of the voicing cues contributes to voicing perception in Japanese is an interesting topic, but it is left for future research.

One interesting aspect of the result is that at least in labials and dorsals, people almost never misperceived voiceless geminates as voiced. On the other hand, voiced geminates, in which internal voicing is weakened, are misperceived as voiceless. This perhaps indicates that Japanese speakers do not hear voicing in geminates unless they are provided with evidence to the contrary.

The difference between coronals on the one hand and labials and dorsals on the other is an interesting one. For labials and dorsals, Japanese speakers are reluctant to hear voiced geminates. Thus, they never hear voiceless ones as voiced, and the correct response to voiced geminates is very low. It thus seems that there is a strong bias against voiced geminates for labials and dorsals. On the other hand, for coronals, Japanese speakers seem to be more confused when faced with a coronal geminates⁷: as a result, the voicing judgment of coronal geminates is much closer to what is expected if people are responding by chance. We can speculate that [dd] is more acceptable for Japanese speakers so that when they listen to a coronal geminate in a noisy environment, they are confused about the voicing quality, so respond more or less by chance, rather than merely rejecting [+voice] perception.⁸

This partially replicates Gelbart's (in prep.) finding that Japanese people are very reluctant to hear a geminate [bb], compared to [dd] (he does not test [gg]). In his experiment, he discovered that given a continuum in which the closure length is varied (with full internal voicing), a labial is not heard as a geminate unless it has very long closure. In fact, even with the longest closure (about 200ms), a labial voiced consonant is not judged to be a geminate 100 percent of the time (but rather a bit above 80 percent). In addition, reaction time for length judgment for labial voiced consonants is systematically longer than that for coronal consonants. This is in line with my result in that there is an additional bias against [bb] (compared to [dd]) in Japanese people's perception.

⁶ No difference is observed between [bb] and [gg]: $F(1, 16)=2.838, p=.111$.

⁷ This is a general pattern, not attributable to a particular token or the combination of a particular file with a particular noise file.

⁸ After the experiment, some subjects did report that coronal geminates are harder to distinguish in terms of voicing quality, compared to others.

The fact that Japanese people do not hear [bb] and [gg] while they are more prone to hear [dd] might suggest that there is an additional grammatical constraint against [bb] and [gg] - [bb] and [gg] are more marked, and hence people are reluctant to hear them. A constraint against [dd] does exist, but the requirement is not very strong. People get confused with [tt] and [dd], rather than rejecting [dd]. That [bb] and [gg] are especially disfavored compared to [dd] is also reflected in frequency of [dd] compared to [bb] and [gg]. According to the database developed by Amano and Kondo (2000), [dd] is much more frequent than [bb] or [gg] (based on tokens; the numbers in parentheses represent the number of types):

(32)

Following Vowel	bb	dd	gg
a	91 (4)	493(14)	142(2)
i	0	13(3)	41(6)
u	398 (6)	0	996(147)
e	0	15(4)	0
o	1 (1)	22375(316)	22(2)
sum	490	22896	1201

The small number of [bb] geminates reflects the fact that a labial often fails to geminate in the right environment (Katayama 1998; Shirai 1999).⁹ To derive this asymmetry, it seems necessary to posit grammatical constraints against [bb] and [gg], which also plays a role in perception, biasing Japanese people against hearing [bb] and [gg].

What is interesting is that this contradicts a purely aerodynamic view of markedness in which it is predicted that [bb] is the least marked among voiced geminates (Hayes and Steraide in press; Kingston 2003; Ohala 1983). The size of oral cavity is biggest for labials, and cheek are susceptible to passive expansion, so keeping voicing should be easiest for labials. Cross-linguistically, it is indeed observed that having [bb] is more frequently than [dd] (see the works cited above). So there must be a Japanese-specific constraint against [bb]. I can only provide a very speculative hypothesis here. Historically, Japanese lost [p] (whose modern reflex is [h]; Ueda 1898), and therefore voicing in labials is not phonemic in the native phonology. So there might be a restriction on having a voicing contrast in labial geminates as well. What is not explained by this hypothesis, however, is that fact the Japanese speakers in my perceptual study did not exhibit any visible bias against a singleton [p], and the same observation was made in Gelbart's experiment, despite the prohibition against singleton [p] in the native vocabulary. Thus, it remains to be seen what the exact nature of a constraint against [bb] is.

On the other hand, the extra markedness of [gg] over [dd] might be attributed to the general marked scale [dd] > [gg], which has an aerodynamic explanation. The size of the oral cavity is bigger for [dd], and [gg]'s capacity to actively and passively expand is much smaller than [dd]'s (see the references above). Therefore, it is easier to maintain voicing in coronals than dorsals, hence the extra bias against [gg].

⁹ [g] is less likely to undergo gemination than [d], although it does not resist gemination as much as [bb].

4. Conclusion

This paper investigated the acoustic and perceptual aspects of voicing in geminates. As Japanese only recently phonemicized voicing in geminates, little work has been done on the interaction of voicing and geminacy. I identified various phonetic cues that signal voicing in Japanese ((i) closure voicing, (ii) closure length, (iii) V1 length (iv) F0 and F1 of surrounding vowels), some of which are weakened in the context of geminates. Most notably, closure voicing, which is arguably the most robust voicing cue (Stevens and Blumstein 1981) is highly weakened in geminates, presumably because maintaining voicing in a long obstruent closure is aerodynamically hard. On the other hand, we have also seen that some cues are enhanced in the context of geminates, perhaps to compensate for the weakening of other cues. Compared to the rather strong and systematic weakening of closure voicing, however, such compensations are subject to inter-speaker variability, and also the effects seem very small. So from an acoustic point of view, it seems likely that voicing cues are weakened in geminates.

This conclusion is supported by the result of the perceptual experiment. It indeed revealed that voicing is much harder to discriminate in geminates in general, supporting Kawahara's (2003) claim. This shows that weakened cues (internal voicing and/or the closure duration difference) constitute important cues for voicing perception in Japanese, and that compensation is not sufficient. What I have not investigated, however, is which acoustic cues contribute to voicing perception in what way. Another question left unaddressed is whether some cues are integrated or not.

Furthermore, I pointed out that there is a possibility that, in addition to weakening of acoustic cues, Japanese might have a perceptual bias against labial and dorsal voiced geminates, which also manifests itself through the patterns of gemination. This explains the especially poor performance of the Japanese speakers in hearing voicing in [bb] and [gg] as well as the fact that Japanese speakers almost never confused voiceless geminates as voiced. This provides another case in which grammatical constraints affect people's perception (Moreton 2002, among others). One remaining question is why [dd] is more favored than [bb], even though [bb] is aerodynamically less marked than [dd].

One general conclusion that can be drawn from the experiments reported in this paper is that phonology is at least partially driven by phonetics. Phonologically, voicing is more easily lost in geminates in Japanese; we observe this asymmetry in the different kinds of response to the OCP(voi). In light of the result of this experiment - that voicing is harder to hear in geminates - we can regard this as another instance in which contrasts signaled by phonetically weak cues are phonologically more prone to neutralization, just as preconsonantal place cues are much more easily lost than prevocalic place cues. This finding adds to the growing body of literature that shows phonological neutralization is closely tied to phonetic perceptibility (Hura et al 1992; Jun in press; Kochetov 1999; Kohler 1990; Ohala 1990; Ohala and Ohala 1993; Steriade 1997, 1999, 2001 among others). This study thus provides an additional endorsement for the claim that phonology is, at least in part, affected by phonetic factors.

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